

SlabMaster™ Enterprise Platform

Operational Architecture & Strategic Roadmap

Version: v1.0.0

Updated: September 2026

Status: Implemented & Production-Ready

1. Executive Overview

SlabMaster™ is an enterprise, multi-tenant cloud platform purpose-built for stone fabrication, quartz manufacturing, and high-volume countertop installation operations. It bridges the critical operational gap between back-office ERP systems (specifically SAP S/4HANA & ECC) and daily shop-floor/field execution—unifying builder accounts, multi-plant manufacturing facilities, direct materials inventory, in-house crews, and third-party installation contractors into a single real-time command center.

2. Core Business Functions & Operational Modules

Builder & Hierarchy Management	Auto-Schedule & Multi-Plant	Inventory & Purchasing	Shop Floor & Field Dispatch
3-Tier Builder Hierarchy - Community & Lot Scoping - Custom Builder Pricing Tiers - Dynamic EAV Attributes	14-Day Visual Dispatch Grid - Configurable Work-Day Engine - Holiday Blackout Logic - Crew Capacity Balancing	- Live Slab & Remnant Tracking - Barcode & QR Label Printing - Vendor Purchase Orders - Receiving Ingestion & Audit	- Touchscreen Kiosk PWA - Station Barcode Advance - Offline Field Lead PWA - Digital Signatures & Photos

A. Three-Tier Account, Community & Lot Hierarchy

- Three-Tier Hierarchy:** Organizes complex commercial and residential production by **Account** (e.g., *Lennar, Toll Brothers, Pulte*) → **Community / Subdivision** → **Specific Lot / Unit Phase**.
- Calculated Milestone Automation:** Forward and backward milestone calculation (Template, CAD Review, Sawjet Cutting, Polish, Quality Inspection, and Final Installation) driven by plant lead-time parameters.

B. Auto-Schedule Engine & Production Calendar

- **Dynamic Work-Day Engine:** Configurable plant production calendars supporting customized work-week schedules (e.g., Mon–Thu 10-hr vs. standard 5-day) and regional holiday shutdowns.
- **Automated Milestone Sequencing:** Lead-time calculation automatically offsets milestone dates based on operational work days, skipping non-working days and plant holiday blackouts.
- **Crew Capacity Balancing:** Real-time load monitoring across plants to prevent over-allocation of in-house install trucks and external contractor crews.

C. Slab & Remnant Inventory Control

- **Full Slab Lifecycle:** Manages stone slabs through AVAILABLE , RESERVED , IN_PRODUCTION , REMNANT , CONSUMED , and DAMAGED .
- **Remnant Monetization:** Automatically provisions remnant offcut records with dimensions and bin/rack allocations for secondary fabrication.
- **QR / Barcode Printing:** Instant thermal label generation for warehouse bin tracking and tablet scanning.

D. Purchasing & Vendor Materials Management

- **Vendor PO Issuance:** Create and track purchase orders for stone distributors (Cosentino, Caesarstone, Arizona Tile) with finish, dimensions, and unit costs.
- **One-Click Receiving:** Warehouse receiving verifies bundle integrity and automatically ingests slabs into live inventory.

E. Table Matrix Form Builder & Custom Attributes

- **Visual Form Matrix:** Design complex tabular forms (Room, Stone Color, Dimensions, Edge Detail, Splash, Sink Cutout, Quantity) with dynamic row calculation.
- **Signature & Photo Capture:** HTML5 touch signature pads for customer sign-offs and camera uploads for installation verification.

F. Shop Floor Kiosk Application

- **Station Touch Interface:** Glove-friendly touch UI for sawjet operators, CNC programmers, edge polishers, and QA staging technicians.
- **Instant Barcode Advancement:** Operators scan barcodes or tap to advance work orders across plant stations, broadcasting progress to the cloud.

G. Mobile Field Dispatch & Offline PWA

- **Role-Tailored Field View:** External 1099 contractors and internal installation trucks receive a streamlined interface displaying only their assigned jobs and required activities.
 - **Data & Pricing Isolation:** Proprietary builder pricing, profit margins, and other subcontractor schedules are strictly shielded from field users.
 - **Offline PWA Engine:** IndexedDB and Service Worker caching allow field technicians to work continuously in basement/concrete dead zones without losing signatures or data.
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3. Enterprise Security, Integration & Resilience Architecture

Pillar	Implementation	Business Value
Authentication (SSO)	Microsoft Entra ID (Azure Active Directory)	Enterprise-grade identity verification using corporate SSO or external Microsoft credentials.
Access Control (RBAC)	Granular Role Matrix (<i>Global Admin, Subscriber Admin, Plant Admin, Estimator, Field Crew</i>)	Principle-of-least-privilege; prevents unauthorized access to pricing, settings, or out-of-scope regional territories.
Two-Way ERP Sync	Modern RESTful JSON API with External ID Routing	Bidirectional synchronization with SAP S/4HANA using external IDs (<code>externalId</code>) across Accounts, Communities, Lots, Jobs, and Activities.
Outbound Resilience	Progressive Retry Queue & Dead-Letter Hub	Exponential backoff ladder (<code>0s → 1m → 5m → 15m → 1h → Dead-Letter</code>) protecting against network glitches and ERP downtime.
Developer Enablement	Postman Collection v2.1 & SAP Developer Pack	Downloadable Postman pack and standalone <code>api-docs.html</code> integration portal with ABAP sample code.
Audit & Compliance	Immutable Change Log	Comprehensive audit logging capturing timestamped modifications, date shifts, user role edits, and price changes.

4. Strategic ROI & Business Impact

- 🛑 **Eliminates Dry Runs & Scheduling Errors:** Synchronizes builder readiness dates with fabrication shop lead times, reducing costly re-trips (\$450–\$750/trip).
- ⚡ **Accelerates Contractor Onboarding:** Self-service Microsoft SSO invitations and automatic plant-territory assignment streamline field partner setup.
- 🔒 **Protects Financial Margins:** Enforces regional pricing rules, isolates proprietary financial data from third-party trades, and ensures complete job traceability.
- 🌐 **Scales Multi-Plant Operations:** Enables enterprise fabricators to expand into new regional hubs while maintaining unified corporate governance and local operational autonomy.
- 🔄 **Automates ERP Synchronization:** Eliminates manual re-keying between SAP and the shop floor, removing human error and accelerating cash-to-cash billing cycles.